Bridging the Gap: Do Better Numerical Embeddings Improve Generation Faithfulness?

The Problem with Numbers in LLMs:

- Arbitrary Tokenization (BPE): Subword tokenizers break numbers into random pieces (e.g., 12345 ⇒ ["12", "345"]).
- **xVal** (Continuous Tokenization):
  - All numbers are mapped to a single placeholder token **[NUM],** then **dynamically scales** by multiplying by actual number.

Other Embedding Approaches

Other paradigms for encoding numbers:

- FoNE (Fourier Number Embeddings): Maps numbers to a single token using **sine/cosine waves** at different frequencies. Extremely effective for high-precision arithmetic.
- Abacus Embeddings: Assigns explicit **positional coordinates to digits relative to their place value** (units, tens, hundreds). Designed to help models handle very long numbers.
- DICE (Deterministic Embeddings): A training-free method where the **cosine similarity between two numeric vectors matches their mathematical distance** on a number line.

Hypothesis

- Numeric Computation ≠ Numeric Generation Faithfulness
- Numerical Computation: The model's ability to compute (e.g., 43 + 27 = 70).
- Numerical Generation Faithfulness: The model's ability to accurately copy or report numbers in generated text (e.g., patient’s temperature is 96 instead of 106, summarizing a report without swapping "4.2M" with "4.2B").
- **The Hypothesis: Improving computation skills (via better embeddings) does not automatically guarantee faithful text generation.**

Why This is Novel

Moving beyond simple arithmetic benchmarks:

- The Gap in Current Literature: Most numerical embedding papers (like xVal or FoNE) only test on **arithmetic or scientific forecasting**.
- Our Target: We test text generation. We want to see if structured number embeddings reduce **numeric hallucinations** in real-world tasks.
- The Benchmark: We will evaluate this on the ToTTo (Table-to-Text) dataset (and many more), checking if the model faithfully describes numerical tables.

Experimental Setup: Isolating number encoding strategy

To prove our hypothesis, we must isolate the embedding strategy completely:

- Two Identical Models: We will train two identical **GPT-2** models from scratch.
- Controlled Variable:
  - Model A (Control): Uses standard BPE for numbers.
  - Model B (Experimental): Uses xVal for numbers.
- Constant Factors: Same training data (**ToTTo** text), same hyperparameters, same non-numeric vocabulary.

How We Will Measure Success

We will evaluate both models on two distinct tracks:

1. **The Computation Track**:
   - Run arithmetic and magnitude probes (e.g., order comparisons, addition, subtraction, etc.).
   - *Metric*: Exact Match Accuracy.
2. **The Generation Track**:
   - Generate descriptions from ToTTo tables.
   - *Metric*: Numeric Token F1, Number-swap detection rates.

Expected Insights

What we'll show:

a. If Model B (xVal) outperforms Model A (BPE) in both computation and generation, **numerical embeddings are a universal fix**.
b. If Model B only improves in computation but still hallucinates numbers in generation, **faithfulness is a decoding/attention problem, not an embedding problem**

- If it's case (a) ⇒ insights about how numerical hallucinations can be mitigated
- If it's case (b) ⇒ propose how numerical hallucinations can be fixed.

Why Train From Scratch?

Why we aren't just fine-tuning:

- **Pre-trained** LLMs already have **BPE-based number representations** in their weights.
- Fine-tuning would introduce confounding variables (pre-training bias vs. new embedding fight).
- *Current progress*: learning how to build and train GPT-2 models.

RESULTS so far (example generation)

BPE-GPT2:

"reference": "Daniel Henry Chamberlain was the 76th Governor of South Carolina from 1874.",

"generated": "Daniel Henry Pud es was the 76 th governor of South Carolina in 1874 .",

Xval-GPT-2:

"reference": "Daniel Henry Chamberlain was the 76th Governor of South Carolina from 1874.",

"generated": "Daniel Henry Archer was a member of the thir teenth Governor of South Carolina from December , to December , , , , , .",